

KATHLEEN YICK

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🌐 kathleen-yick.github.io/maker_portfolio

Education

Massachusetts Institute of Technology

Sep. 2025 – May 2029

Bachelor of Science in Mechanical Engineering

Cambridge, MA

- Relevant Coursework: Mechanics and Materials, Differential Equations, Solid State Chemistry, Intro to Python and Programming, Electromagnetism
- Activities and Societies: Assistive Technology (MechE Lead), MIT Makerspaces (Mentor), Technique Yearbook Club

Experience

MIT Scrap Intelligence Lab

June 2026 – August 2026

Mechanical Engineering Student Researcher

Cambridge, MA

- Designed and optimized the Magnetic Induction Spectroscopy (MIS) sensing subsystem as part of a multidisciplinary team developing an AI-powered scanner for rapid nonferrous scrap metal identification.
- Integrated electromagnetic sensing, computer vision, weight sensing, and machine learning into a multi-sensor prototype capable of classifying copper grades and detecting contaminants in mixed scrap streams.
- Owned the mechanical design and fabrication of the data acquisition platform, integrating load cells, electromagnetic sensors, and structural components for repeatable testing.
- Conducted 20+ customer discovery interviews with scrapyards owners, recyclers, and metal processors to translate operational challenges into engineering requirements for a deployable industrial sensing platform.

MIT Leap Group

January 2026 – May 2026

Mechanical Engineering Student Researcher

Cambridge, MA

- Designed and evaluated experimental protocols to assess user interaction with a 7th Axis Universal Robot within a ROS-based teleoperation framework.
- Collaborated with graduate researchers to test sensor-guided pick-and-place workflows and collect user feedback for accessibility-focused manufacturing research.
- Recommissioned an ABB YuMi IRB 14000 by troubleshooting hardware/software issues to restore for laboratory testing.

Hideo Sasaki Foundation

2024 – 2025

Architectural Intern

Chinatown, MA

- Designed a conceptual plan for a 7-acre brownfield redevelopment into a multi-use public park incorporating flood mitigation features and 20+ native plant species to improve ecological resilience.
- Conducted traffic pattern analysis and developed proposed roadway layouts and signage modifications for the City of Malden, with recommendations projected to reduce parking exit conflicts by 50%.

Projects

Blue Origin x Momentum MIT Innovation Challenge | *Systems Engineer*

January 2026

- Led systems-level design of a self-sufficient Mars colony integrating orbital mechanics, propulsion systems, and AI.
- Awarded 1st place by a panel of Blue Origin engineers.

Modifying Standard-Grade Rollator to Optimize Self-Autonomy | *Team Lead* September 2025 - December 2025

- Applied Fusion 360, FEA, and MIG welding to modify an industry-standard rollator for a co-designer with unique needs.
- Redesigned foldable back support for increased surface area and grip, enhancing user stability and ergonomics.
- Engineered a reinforced, high-friction seat to safely support weight while improving comfort and usability.

Chinatown Boston Public Library Branch Greenspace | *Architectural Intern*

January 2025 - May 2025

- Redesigned an underutilized parking lot adjacent to the new BPL branch into a community greenspace addressing Chinatown's deficit in public open space and increasing youth engagement by 25%.
- Developed phased programming strategies with the city: library services, youth engagement, and revenue-offset solutions.

Technical Skills

Software: CAD (Rhino, Fusion360, OnShape), FEA (Fusion360), Java, Python, ROS (Python), Adobe Suite (Photoshop, Illustrator, InDesign), Figma

Hardware: 3D Printing, Laser Cutters (Metal, Wood), Manual Milling, Lathe, Metalworking, MIG Welding

Interests: Dressmaking, Rock climbing, Tennis, Photography, and Metal Origami, Drawing